

Forward calculation of gravity and its gradient using polyhedral representation of density interfaces: an application of spherical or ellipsoidal topographic gravity effect

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Abstract A density interface modeling method using polyhedral representation is proposed to construct 3-D models of spherical or ellipsoidal interfaces such as the terrain surface of the Earth and applied to forward calculating gravity effect of topography and bathymetry for regional or global applications. The method utilizes triangular facets to fit undulation of the target interface. The model maintains almost equal accuracy and resolution at different locations of the globe. Meanwhile, the exterior gravitational field of the model, including its gravity and gravity gradients, is obtained simultaneously using analytic solutions. Additionally, considering the effect of distant relief, an adaptive computation process is introduced to reduce the computational burden. Then features and errors of the method are analyzed. Subsequently, the method is applied to an area for the ellipsoidal Bouguer shell correction as an example and the result is compared to existing methods, which shows our method provides high accuracy and great computational efficiency. Suggestions for further developments and conclusions are drawn at last.

Keywords Density interface · Polyhedron · Gravity · Gravity gradient · Icosahedron

1 Introduction

The accurate and efficient calculation of gravity and its gradient data of density interfaces, the Earth's undulant surface for instance, has been studied by numerous researchers over the last several decades. In which forward modeling the topographic gravity effect and removing it from the gravity observation is commonly known as the terrain correction. (e.g., [Li and Sideris 1994](#); [Chapin 1996](#); [Feather and Dentith 1997](#); [Leaman 1998](#); [Nowell 1999](#)). The process is essential for utilizing gravimetric technique to study the physical shape and also interior density contrast structure of the Earth and other planets. [Bullard \(1936\)](#) proposed the Bullard-B correction to reduce the gravity effect of the Earth's curvature in the local gravity survey. It is later modified and observed by [Lafehr \(1991b, 1992\)](#), [Hensel \(1992\)](#) and [Talwani \(1998\)](#). In general, the Bullard-B correction is enough for local studies. However, for regional or global studies that stretch over large geographic regions, spherical or ellipsoidal terrain correction is needed, especially over mountainous areas with drastic elevation changes ([Lafehr 1998](#); [An et al. 2015](#)). Existing methods for forward calculation of a density interface's gravitational field that discussed in the literature cover both spatial and spherical harmonic domains. [Wieczorek and Phillips \(1998\)](#) improved Parker's formula ([Parker 1973, 1975](#)) from the Cartesian frequency domain into the spherical harmonic domain and applied it to the global lunar gravity field. [Balmino et al. \(2011\)](#), [Hirt et al. \(2012\)](#) and [Bouman et al. \(2013\)](#) introduced the spherical harmonic method to calculate the Bouguer and isostatic anomalies for satellite gravity and its gradient fields of the Earth. On the other hand, spatial approaches usually divide the Earth's surface into small regions with regular blocks such as the spherical prism, and then the gravitational effect of the undulant interface is approached as a cumulative effect of all blocks (e.g., [Ardalan](#)

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and Safari 2004; Claessens and Hirt 2013). Heck and Seitz (2007) and Wild-Pfeiffer (2008) used the Taylor series to solve the integration of the gravitational field generated by the spherical prism. Later, the Gauss–Legendre quadrature integration is applied to the problem by Asgharzadeh et al. (2007) and Grombein et al. (2013). Subsequently, the integration is applied to the gravity field of an ellipsoidal prism by Roussel et al. (2015). Another branch of spatial approaches is using polyhedrons to approximate the surface undulation. Werner and Scheeres (1996) proposed a forward algorithm for the gravitational field of a homogeneous polyhedral source and applied the method to evaluate the gravitational field of an asteroid. Tsoulis et al. (2009b) introduced the computation of the potential harmonic coefficients of a constant density polyhedron. And analytical solutions of full gravity tensor of a homogeneous arbitrarily shaped polyhedral source are derived by Tsoulis (2012) using line integrals.

There are some disadvantages in each of these previous methods. Eshagh (2013) pointed out that, due to a term of $(r/R)^n$ in the spherical harmonic expression, the spherical harmonic-based methods are unstable in high spherical harmonic degrees n when geocentric radius of the Earth's terrain r is larger than its mean radius R . Meanwhile, spatial approaches that use numerical integration techniques have variable accuracies that change with the distance between the source and the observation point (Werner and Scheeres 1996). The computation is also usually time consuming and even singular when the observation point is located very close to the source (Wild-Pfeiffer 2008; Uieda et al. 2011). And special treatments may need to be applied to eliminate singularities while using polyhedral sources (Tsoulis and Petrović 2001; Urso 2013). Moreover, most forward modeling methods adopt geographic coordinates as their basis because this is commonly used in a wide array of data sets and processing algorithms. However, the use of geographic coordinates introduces definite limitations (Sahr et al. 2003). The actual size and shape of grid cells under the latitude–longitude graticule vary along with the cell's latitude, and they become increased distorted as one moves north and south from the equator. Therefore, methods that are defined under the geographic framework also suffer for this disadvantage, especially when one deals with polar regions.

To address deficiencies of former methods, in this paper, we propose a spatial approach to construct the 3-D spherical or ellipsoidal density interface model and forward calculation of its gravitational field. The approach mainly includes a new density interface modeling method that is referred to as the Polyhedral Global Interface Model (PGIM) and an adaptive forward calculating process, called the Near Zone Replacement (NZR), that enhances the computational efficiency. Firstly, we introduce the methodologies of the PGIM and NZR and analyze their unique characteristics. Subsequently, the proposed method is applied to a selected area to

perform the Bouguer shell correction. Suggestions for further improvements and general conclusions are drawn at last.

2 Methodology

2.1 Global interface modeling with the PGIM

In this section, we will introduce a spatial approach for constructing global density interface models with polyhedrons, which is called as the Polyhedral Global Interface Model (PGIM). This approach is developed deriving from the Geodesic Discrete Global Grid Systems (Geodesic DGGS) (e.g., Fuller 1975; Sahr et al. 2003). In general, unlike grids based on geographic coordinates, the Geodesic DGGS employs some spherical polygons, such as the spherical triangle and hexagon, as topologically equivalent surrogates for the Earth's surface. The main advantage of the Geodesic DGGS is that it provides nearly constant resolution all over the globe (Sahr et al. 2003). Similar to the Geodesic DGGS, several steps are followed to construct the PGIM.

- (a) Choosing a regular base polyhedron;
- (b) Determining a fixed orientation of the base regular polyhedron relative to the spherical coordinate system;
- (c) Using a hierarchical spatial partitioning method defined symmetrically on a face (or set of faces) of the base regular polyhedron to divide the polyhedral surface into small facets;
- (d) Determining geocentric radiuses of vertices of the partition after they are mapped to the corresponding interface's surface;
- (e) Inversely mapping vertices of the partition to the interface's surface to construct 3-D models.

Firstly, for choosing a base regular polyhedron, we adopted an icosahedron as the basis to construct the PGIM (Fig. 1a). As discussed by White et al. (1992, 1998) and Sahr et al. (2003), a Geodesic DGGS developed on the icosahedron divides the spherical surface into same spherical triangular regions, which are topologically inclined for undulant surface modeling, that has six triangles meeting at each vertex, with the exception of the 12 original vertices of the icosahedron that are connected to only five triangles. In addition, as noted by White et al. (1998), the distortion introduced by transforming between a face of a platonic solid and its outer spherical surface is smaller when the platonic solid has smaller faces. Therefore, comparing to other platonic solids, the icosahedron introduces minimal distortion while mapping its faces to the corresponding outer spherical surface. Consequently, differences of size and shape between triangular regions after the transformation are minimized as well,

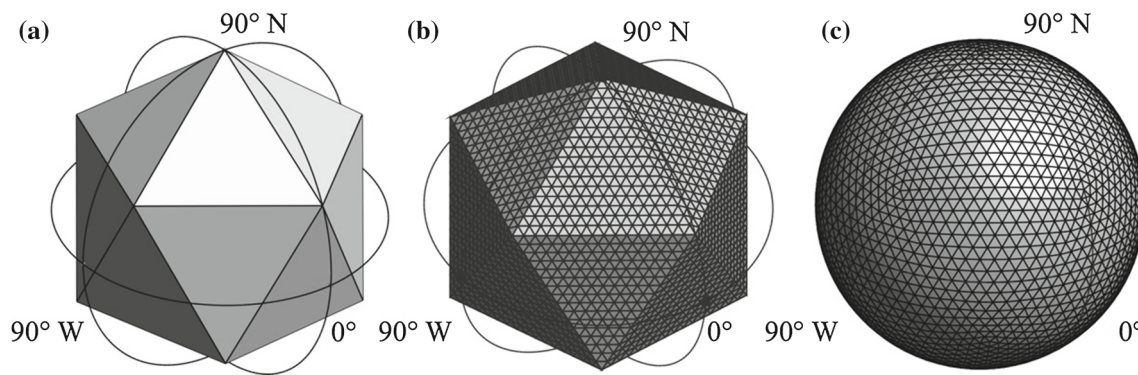


Fig. 1 Process of construction of the PGIM. **a** A diagrammatic sketch of an icosahedron. One vertex of the icosahedron locates at each of the poles and one of the edges aligns with the prime meridian. **b** Triangular

subdivision of the icosahedral faces in **(a)**. **c** An approximation of a sphere by mapping the partition of the icosahedron in **(b)** to an outer spherical surface

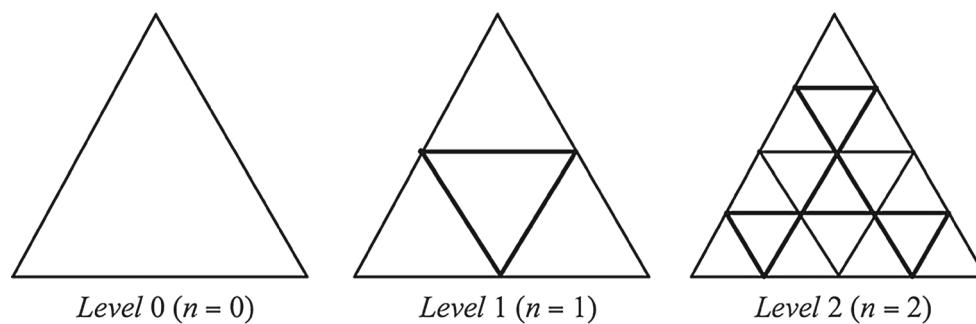


Fig. 2 Three levels of a *Class I* recursive subdivision defined on a triangular facet, n represents times of recursive subdivision. Practically this strategy could be performed using a triangular grid quad tree algorithm (Fekete and Treinish 1990; Dutton 1999)

which enables the resulting grid to maintain nearly constant fineness across the globe (Gregory 1999). As for the orientation of the icosahedron, the most common choice, adopted by Williamson (1968), Sadourny et al. (1968), Fekete and Treinish (1990) and Thuburn (1997), is to place the icosahedron in the spherical coordinate with a vertex placed at each of the poles and one of the edges aligned with the prime meridian (Fig. 1a). Because the icosahedron is used as a skeleton to partition the spherical space and then vertices of the partition are mapped to an outer surface, the initial size of the icosahedron is not crucial. The length of the icosahedron is set to a default value $l = 1e + 5$ m in this paper.

Subsequently, to partition all 20 faces of the icosahedron into sets of smaller triangles, we employ the method of Fekete and Treinish (1990) and White et al. (1998) that is called the *Class I* or alternate subdivision (Kenner 1976). Each face of the icosahedron is divided into four smaller equilateral triangles by breaking each edge into two pieces at the middle point and connecting the break points with lines parallel to the triangular edges (Fig. 2). Recursively subdividing the resulting triangles divides the surface of the icosahedron into a set of partitions which are organized hierarchically by the times of subdivision (Dutton 1999). Given the subdivision

has been taken n times, then the number of triangular facets of a level n subdivision of the icosahedron equals 20×4^n , in which the level 0 partition represents the base icosahedron (Fig. 1a), while a higher level represents a finer partition of the icosahedral surface (Fig. 1b).

Subsequently, in order to model an undulant interface such as the Earth's surface based on the partition preformed on the icosahedron, we need to map all vertices of the partition to the interface's surface according to their geocentric radiuses on the surface. However, longitudinal and latitudinal coordinates of these vertices mostly do not coincide with grid nodes of an equal-angle grid-based digital elevation data set such as the ETOPO1 (Amante and Eakins 2009) (Fig. 3), which means geocentric radiuses of the PGIM's vertices on the interface usually cannot be obtained directly. Therefore, we firstly calculate geocentric radiuses of the interface under an equal angular grid, then final radiuses of the vertices are determined using a spherical bilinear interpolation method proposed by Wang et al. (2006), which is introduced in "Appendix A".

Following the steps described above, we could build 3-D digital models of spherical or ellipsoidal interfaces that are designed for forward modeling of gravitational fields (Fig. 4).

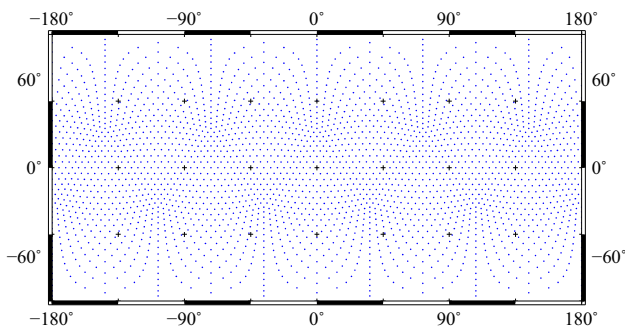


Fig. 3 Node positions of a level 4 PGIM's vertices in the geographic coordinates. It is clear that for most vertices of the PGIM, their coordinates do not coincide with grid nodes of the geographic coordinate

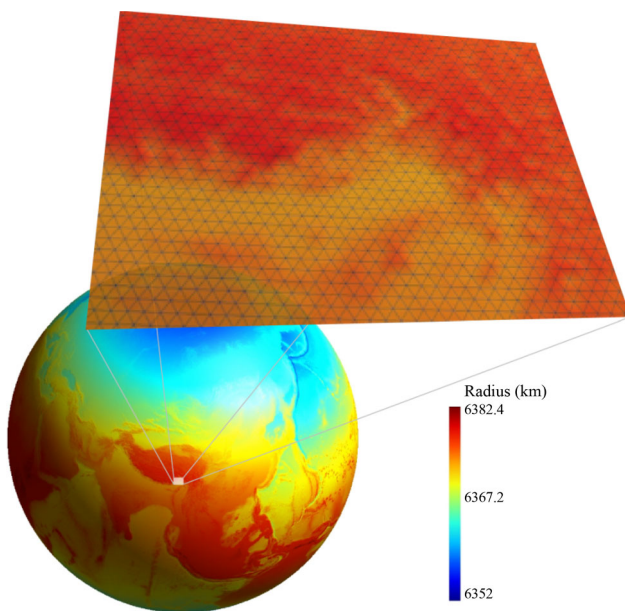


Fig. 4 Image of a level 8 PGIM (the times of subdivision equals 8) of the Earth's surface. The upper enlarged picture shows actual structures of the model

Different models could be easily developed by defining different data sets of interfaces and repeating the last two steps. Subsequently, the exterior gravitational field of the PGIM is achieved by adopting a forward method proposed by [Werner and Scheeres \(1996\)](#) (see "Appendix B").

2.2 Near zone replacement

As noted by [Mikuška et al. \(2006\)](#) and [Lafehr \(1991a\)](#), a prominent feature of the spherical or ellipsoidal topographic gravity effect is that the surface undulation near an observation point produces significant impacts on the gravity and gravity gradient data, and by contrast, the distant undulation has weaker effects. This feature emphasizes the importance of high resolution and accuracy of the surface undulation in

the nearby area of an observation point. Consequently, proper reduction of the accuracy and resolution of distant surface undulation would not result in large errors of the resulting gravity data ([Tsoulis 1998](#); [Tsoulis et al. 2009a](#) and [Uieda et al. 2015](#)). On the other hand, speaking in terms of computational efficiency, decreasing distant surface resolution could also reduce the computing burden since the amount of calculation shrinks.

To apply this conception to the PGIM, the process is presented as following. Given a level n PGIM (n times subdivision) and the surface undulation inside a nearby area of a point P is shown in Fig. 5a. The near zone of P is controlled by a threshold value Θ_P . If the spherical angle θ_P between P and the center of a triangular face is smaller than Θ_P , then the face is assigned as a part of the local surface representation of P ; otherwise, it is considered as a part of the regional or global model of P . To improve the resolution of the local representation, a higher level of subdivision (e.g., $n + 1$ or $n + 2$) is implemented to the local model (Fig. 5b). In which case, the nearby area is replaced by a finer partition. Therefore, this strategy is referred to as the near zone replacement (NZR). Consequently, within the area near P , we use a fine surface representation to get accurate details of the gravitational field, and a rougher global model which provides regional variations and mean amplitude levels. Practically, the NZR could be performed by firstly forward calculating the gravity effect g_{glo} of a level n PGIM at an observation point P (Fig. 6a), then determine the near zone of P and develop a higher resolution of it (Fig. 6b). After that, calculating the gravity effects g_{ori} and g_{imp} caused by the original and finer local models. Subsequently, the gravity effect of the PGIM with the NZR at P is obtained as a cumulative effect $g_{NZR} = g_{glo} + g_{imp} - g_{ori}$ (Fig. 6c).

3 Analysis of the PGIM

3.1 Spatial resolution of the PGIM

In order to utilizing the PGIM to forward modeling gravitational fields, we firstly need to assess the spatial resolution of the PGIM. As it consists of only triangular facets and their vertices do not coincide with the regular equal angular grid, the resolution cannot be evaluated directly. Alternately, we firstly build a level 8 PGIM of the Earth's reference sphere with the radius $r = 6371.393$ km. Histograms of areas, edge lengths and minimum interior angles of triangular faces of the PGIM are shown in Fig. 7. Blue blocks in the figure indicate ranges up and down to 10% of the mean values, which suggests that most of triangular facets are closely equal in terms of areas and edge lengths. Meanwhile, the minimal interior angle of the facets is greater than 54° , so all facets are nearly equilateral. The results indicate that the PGIM is

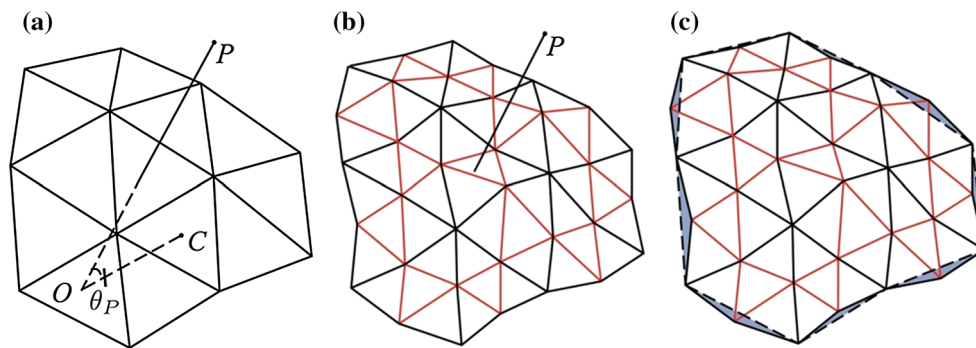


Fig. 5 Near zone replacement (NZR) of an observation point P . **a** A local surface representation of P that contains triangular faces within the range $\theta_P < \Theta_P$. The rest of the PGIM is not shown here. **b** A finer local model of P . Each triangular facet in **(a)** is divided into four

smaller ones. New edges are drawn in red. **c** Gaps, shown as blue triangles, exist at the edge of the near zone between models in **(a)** and **(b)**

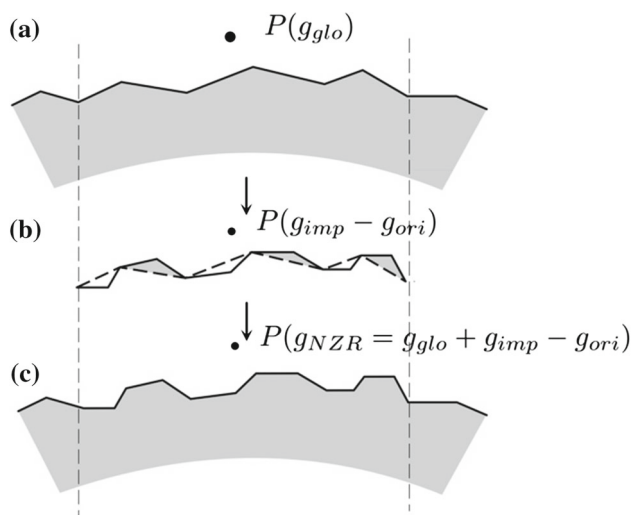


Fig. 6 Integration of the gravity effect of a finer local surface representation into the global model. **a** Gravity observation g_{glo} at the point P caused by the whole PGIM. **b** Difference of the gravity observation $g_{imp} - g_{ori}$ between the original model (dashed line) and improved model (solid line). **c** Integration of gravity observation at P with improved local representation is obtained by $g_{NZR} = g_{glo} + g_{imp} - g_{ori}$

approximated equal resolution across the globe (Fekete and Treinish 1990; White et al. 1998).

Subsequently, a set of PGIMs of the Earth's reference sphere from level 6 to level 10 are used to assess the PGIM's resolution. Table 1 shows statistics of triangular faces' area, edge length and the equivalent angle which represents a geocentric angle of the average edge length with respect to the Earth's reference radius. If taking the angle as a measurement of resolution, then we can observe that the resolution of the PGIM increases from $1^{\circ}4'12''$ (1.07°) to $4'12''$ (0.07°), and every time n increases, the resolution increases nearly in double. Meanwhile, we notice that triangular faces cover smaller areas than squares with similar edge lengths, which, in fact,

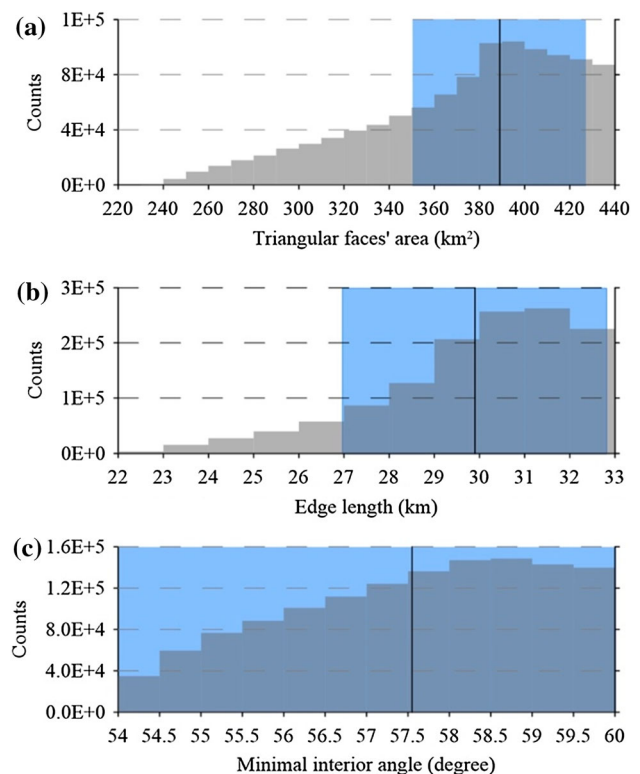


Fig. 7 Histograms of facet's parameters of a level 8 PGIM of the Earth's reference sphere with radius $r = 6371.393$ km. **a**, **b**, **c** Distributions of areas, edge lengths and the minimum interior angles of triangular facets of the PGIM, respectively. The black vertical lines represent the mean values of the corresponding parameters and the blue blocks highlight the range of up and down 10% of mean values

provides better resolution. Therefore, taking both the average area and edge length into consideration, the resolution of a level 6 reconstructed terrain model is given as 1.0° qualitatively, and the resolution increases doubly every time n increases.

Table 1 Qualitative resolution of the PGIMs of the Earth's reference sphere

n	Number of faces (20×4^n)	Average area (km^2)	Average edge length (km)	Equivalent angle (dd mm ss)
6	81,920	6226.67	119.77	1°4'12"
7	327,680	1556.76	59.89	32'24"
8	1,310,720	389.19	29.94	16'12"
9	5,242,880	97.29	14.97	7'48"
10	20,971,520	24.32	7.48	4'12"

3.2 Modeling errors

One important feature of using the PGIM to approximate a spherical interface like the terrain undulation of the Earth's surface is the curvature of the sphere or ellipsoid should be considered first (Çavsak 2012; Lane 2009). In the case where there are no terrain and geoid undulation of the Earth's surface, the PGIM still needs to fit the curvature of Earth's reference sphere or ellipsoid. In order to evaluate the difference between the PGIM and the Earth's reference sphere in terms of their exterior gravitational fields, a synthetic example is given using level 7–10 PGIMs of the Earth's reference sphere with radius $r = 6371.393$ km. Given the densities of the PGIMs are 5.52 g/cm^3 (mean density of the Earth) and the observation points are located 10 km above the reference sphere. Differences between the gravity data produced by the PGIMs and the theoretical value are shown in Fig. 8. In which the theoretical value represents the exterior gravity of the Earth as a solid sphere with the same radius and density as the PGIMs, which is about $979,828.494 \text{ mGal}$. Taking from Fig. 8, differences are larger at centers of the icosahedron's faces and smaller near the original vertices of the icosahedron. Because of varying distances from the partition's vertices to the outer spherical surface, different distortions of the triangular faces are introduced while mapping the partition on icosahedron to the outer spherical surface (White et al. 1998). This feature indicates the PGIM fits better to the curved surface near the original vertices of the icosahedron than at centers of its faces. That is to say the resolution of the PGIM is relatively better near the original vertices of the icosahedron. Meanwhile, the finer resolution the PGIM has, the smaller difference is between exterior gravity of the PGIM and the Earth's reference sphere. For instance, the level 10 PGIM of the Earth's reference sphere produces approximately 0.5 mGal error comparing to the theoretical value, which is adequate for global studies. Consequently, for utilizing the PGIM to calculate gravity effects of the Earth, differences between the PGIM and the Earth's curvature could be ignored if the level of the PGIM is big enough. Similarly, the error of reconstruction an undulant surface will be quite small if we use a high level PGIM.

Besides the difference between the PGIM and the curved surface, another error of calculating the PGIM's exterior gravitational field is introduced by implementing the NZR to the forward calculating process. One feature of replacing an observation point's local surface representation using a finer partition is that it will leave gaps at the edge of near zone. Therefore, the surface of the PGIM is no longer closed (Fig. 5c). This feature introduces some errors into the result of the PGIM's gravitational field. Taking the effect of distant terrain relief into account, the error might be ignored as the threshold value Θ_P increases. Because the gap size is related to the PGIM's fineness as well as the roughness of local terrain surface, the determination of the threshold value Θ_P will be discussed in detail in the following section.

4 Application

In this section, we apply the PGIM to an ellipsoidal Bouguer shell correction (Chapin 1996) of gravity and its gradient data within the area from 60°E to 120°E , and 0°N to 60°N (Figs. 12, 13). This area contains both the Tibetan Plateau and part of the Indian Ocean. The terrain undulation within the area is almost the most drastic across the Earth, which makes it very suitable to test our method. To use the PGIM to perform the Bouguer shell correction, we construct three PGIMs that are built with specific terrain surfaces and densities (Fig. 9). The topographic and bathymetrical gravity effect is achieved as a cumulative effect of these three PGIMs' gravitational fields. Meanwhile, additional factors such as the atmosphere, ice caps, lakes, reservoir rivers, closed seas, as well as the vertical variation of density in the oceanic water are not further considered in this application for simplicity (Eshagh and Sjöberg 2009; Sjöberg 2001; Tenzer et al. 2008). The geocentric radiuses of the PGIMs are determined using the WGS84 reference ellipsoid (e.g., Fairhead et al. 2003; Li and Götze 2001), the ETOPO1 Bed surface global topography and bathymetry data (Amante and Eakins 2009) and EGM96 (Lemoine et al. 1998) global geoid model. The computation points are set 10 km above the WGS84 reference ellipsoid.

Before the correction is applied to the region, a typical profile along 85°E from 15°N to 30°N (Fig. 10a) is firstly

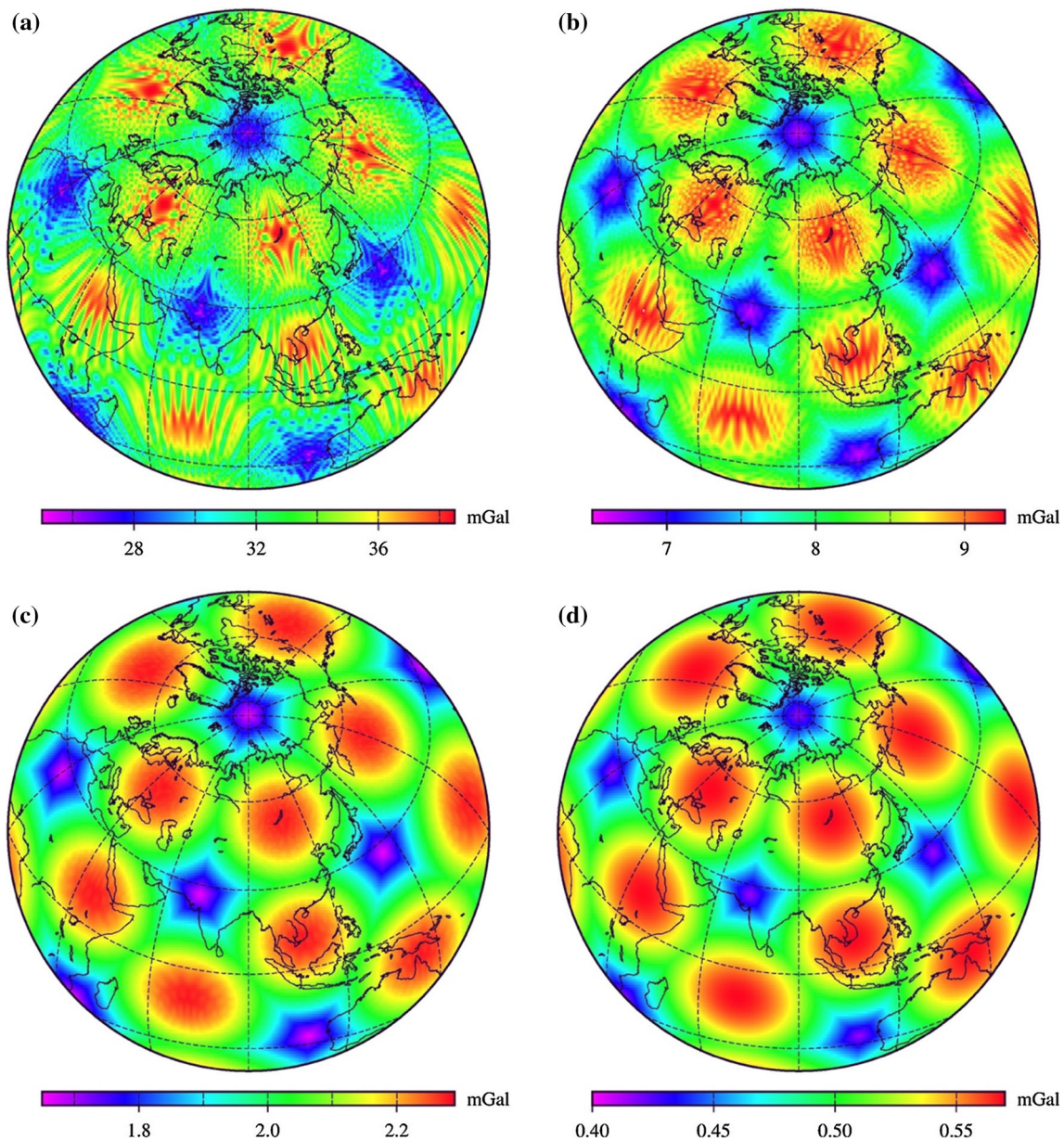


Fig. 8 Differences of the PGIMs' gravity field to the theoretical value which equals 979,828.494 mGal. The subdivision times of the PGIMs of the Earth's reference sphere in (a) to (d) are from $n = \{7, \dots, 10\}$.

Radius of the reference sphere is 6371.393 km; the density is given as the Earth's mean density $\rho = 5.52 \text{ g/cm}^3$ and the observation points P lie 10 km above the reference sphere

investigated in order to determine related parameters, such as the level of regional/global and local terrain undulation models and threshold value Θ_P of the NZR. Drawn from previous sections, we choose a level 10 PGIM to construct local terrain undulations for all observation points. A level 7 and level 8 PGIM are employed to construct the regional/global terrain models, respectively. Additionally, different threshold values of $\Theta_P = (5^\circ, 10^\circ, 20^\circ)$ are used to calculate the terrain correction of gravity V_r and its gradient V_{rr} , $V_{r\theta}$ and $V_{r\phi}$ along the profile with a 0.05° spacing. Meanwhile, gravitational data obtained using the level 10 PGIM without the

NZR is taken as the standard for comparisons (Fig. 10d–g). Figures suggest that errors caused by the NZR become almost undetectable when $\Theta_P = 20^\circ$ and the regional/global and local models are built with the level 8 and 10 PGIM, respectively. The total run time is about 3046s without the NZR and 752s with it on our workstation (Intel Xeon E5, 64 Gigabytes Memory), which is a factor of 4 difference. Therefore, it is obvious that the NZR improves the computational efficiency significantly.

Meanwhile, we compare our result to the ones obtained using the method proposed by Du et al. (2012) (Fig. 11b).

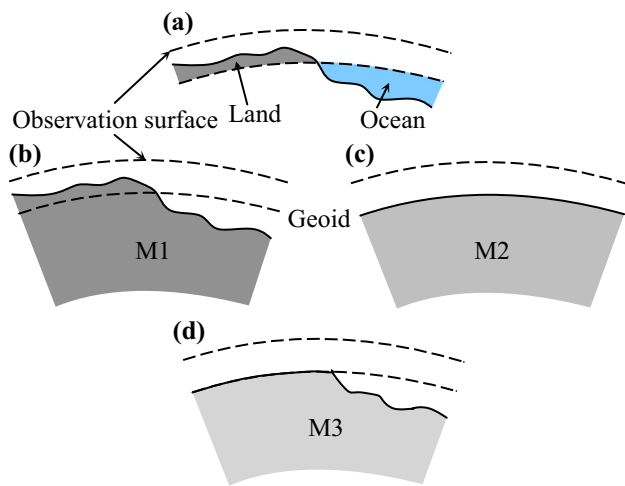


Fig. 9 Methodology and model settings for utilizing the PGIM to perform the Bouguer shell correction. **a** Land mass and ocean that gravity effects need to be removed and modified in the Bouguer shell correction. The boundaries between the land mass and ocean are determined by the elevation data with respect to the Earth's geoid. The densities of the land mass and ocean are given as 2.67 and 1.03 g/cm³, respectively. The observation surface lies 10 km above the reference ellipsoid. **b, c, d** Three PGIMs constructed to perform the Bouguer shell correction. The surfaces of model M1 and M2 coincide with the topography and bathymetry data and the Earth's geoid, respectively. The surface of M3 coincides with the Earth's geoid over the continental area and the bathymetry data over the oceanic area. The densities of M1, M2 and M3 equal 2.67, 1.64 and 1.03 g/cm³. Therefore, the Bouguer shell correction is obtained by a cumulative effect of the models' gravitational fields as $V_{tc} = v_{M1} - v_{M2} - v_{M3}$

Comparison shows the difference on the correction of gravity data ranges from -4 to 1 mGal over the Indian Ocean and the Indian Peninsula, where terrain changes mildly, and the difference increases to -11 to 5 mGal over the Tibet Plateau (Fig. 11c). The greater difference may be the result of larger differences of the terrain models used by different methods due to drastic changes of terrain at the Tibet Plateau, or it could be caused by the shorter distance from the Tibet Plateau to the observation surface, compared to other places. The difference on the correction of vertical gradient data shows the same pattern, which is about ± 1 E over the low altitude area and ranges from -25 E to 18 E at the Tibet Plateau (Fig. 11c). Overall, differences between two methods are minute. And the precision is adequate for regional or global studies. Additionally, the run time of the method given by Du et al. (2012) is about 2872 s, which is a little less than the time as our method without the NZR.

Taking from results above, we utilize same parameters ($\Theta_P = 20^\circ$, a level 8 PGIM for regional/global terrain model and a level 10 PGIM for local terrain representation) for regional application (Fig. 12a–d). Then the Bouguer gravity anomaly of the region (Fig. 13b) is calculated with the gravity disturbance field obtained from the EIGEN-6C4 global gravity model (Förste et al. 2014) (Fig. 13a). The field is

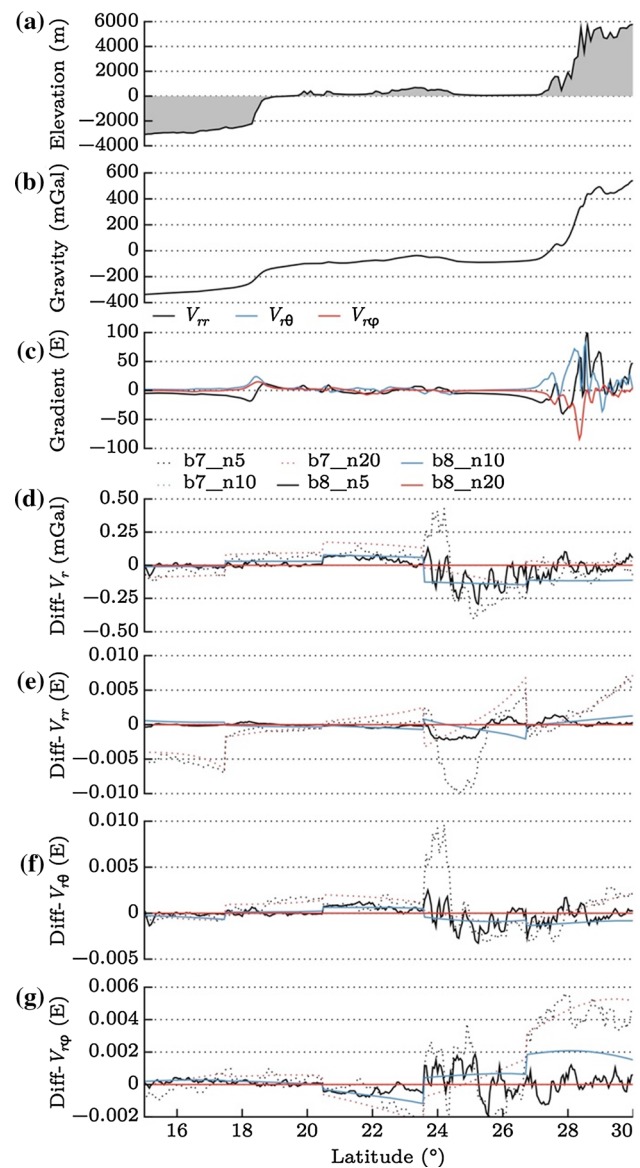


Fig. 10 Features of the NZR along the profile at 85°E from 15°N to 30°N . **a** Topography and bathymetry along the profile. **b** gravity V_r , **c** radial gravity gradient V_{rr} , longitudinal gravity gradient $V_{r\theta}$ and latitudinal gravity gradient $V_{r\phi}$ data of the topography and bathymetry along the profile obtained without the NZR, these are used as standard for comparison. **d, e, f, g** Differences of the gravity and gravity gradient data obtained with different parameters combinations of the NZR comparing to results in (b) and (c). For different parameters combinations, the abbreviations of the legend represent the level of the base models and the range of the NZR. For instance, b7_n5 means a level 7 PGIM as the global model and the range of the NZR is 5°

completed up to spherical harmonic degree/order 2190. The Bouguer gravity anomaly of the region calculated by the ICGEM's online calculation service is also given (<http://icgem.gfz-potsdam.de/ICGEM/Service.html>) here for comparison (Fig. 13c). Having similar patterns of the Bouguer anomalies in Fig. 13b, c, the anomaly obtained with the ICGEM's service shows intense negative anomalies over the

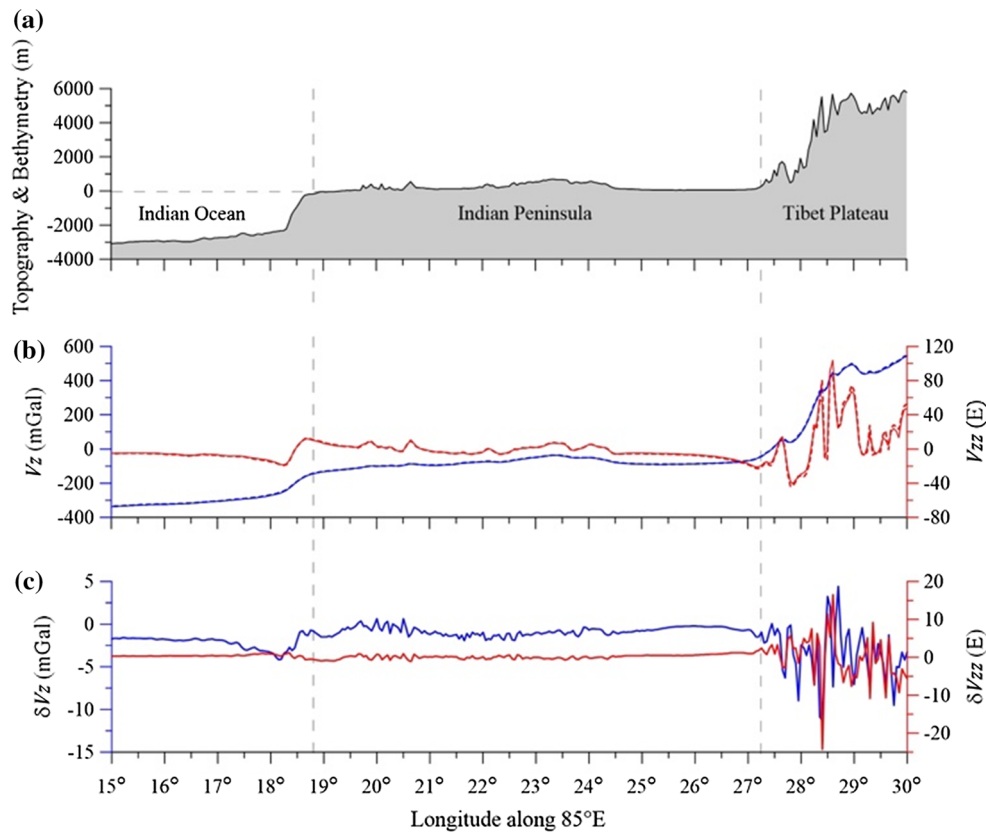


Fig. 11 **a** Topography and bathymetry along the profile at 85°E. **b** gravity and vertical gradient data of the topography and bathymetry along the profile at 85°E obtained by our method (solid lines) and the

algorithm proposed by Du et al. (2012) (dotted). **c** Differences of gravity and vertical gradient between the figures in (b)

continent. The maximal amplitude is about -724.58 mGal found at the Himalaya. The zero anomalies lie along the coast areas. The ocean area in Fig. 13c shows moderate positive gravity anomalies, and the maximal amplitude is about $+365.11$ mGal. As the ICGEM's service uses the Bouguer plate method for the spherical terrain correction, the Earth's curvature is ignored. In contrast, the Bouguer anomalies obtained with our method shows more balanced amplitudes between positive anomalies over the ocean and negative ones over the continent, comparing to the figures in Fig. 13c. The amplitude range in Fig. 13b is from -496.47 to $+502.49$ mGal. Areas with low to moderate altitudes, such as the eastern China and India, show relative weak positive anomalies, from 0 to 100 mGal, which could be interpreted as influences from the mass deficiency of the ocean. In this case, Fig. 13b gives a more realistic view of the gravity effect of Earth's interior density contrasts.

5 Conclusions

We proposed an approach to construct the 3-D density interface model and forward calculating its gravitational fields.

The method is applied to calculate spherical or ellipsoidal topographic gravity effect and performed a Bouguer shell correction as an example. The outstanding advantages of our method are as follows:

- The interface's surface is approximated using triangular facets, and the model's resolution is nearly constant at different locations of the globe. The subdividing routine of the icosahedral facets is defined hierarchically, and thus different resolutions of the model are accessible easily.
- The full gravitational field outside the PGIM, from the gravitational potential down to its gradient tensors, is forward calculated using exact analytic solutions without singularity. The kernel matrixes of the forward algorithm are uniform for all gravitational components, which mean that multiple components could be obtained without increasing much computing burden. Meanwhile, the forward algorithm is globally uniform and no splicing is required. It could be applied to terrestrial data on land, on the sea and even on the seafloor, airborne data and satellite data, etc.

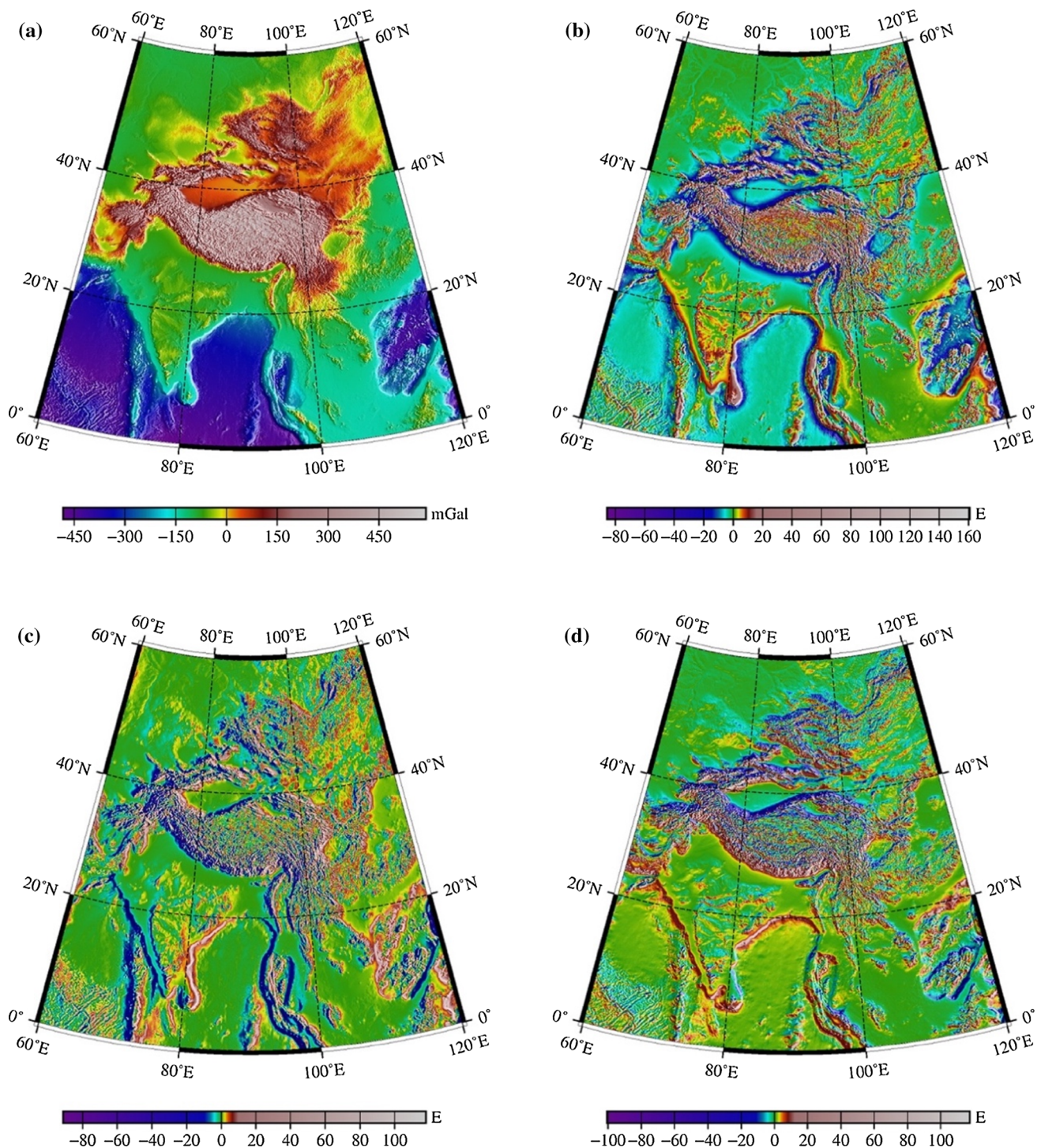


Fig. 12 **a** Gravity V_r , **b** radial gravity gradient V_{rr} , **c** longitudinal gravity gradient $V_{r\phi}$ and **d** latitudinal gravity gradient $V_{r\theta}$ data of topography and bathymetry over the application area

(c) The total error of the PGIM mainly comes from the original elevation data sets. The error introduced by using the planar triangles to approximate the spherical curvature and the original interface undulation could be ignored. Moreover, no large error will be introduced into the gravitational data given a large enough range of the near

zone of the NZR. And for those applications that require ultra-high precision of the forward fields, we recommend using the PGIM without the near zone replacing process.

(d) Other than calculating topographic gravity effect, the method is suitable to reconstruct density interfaces of

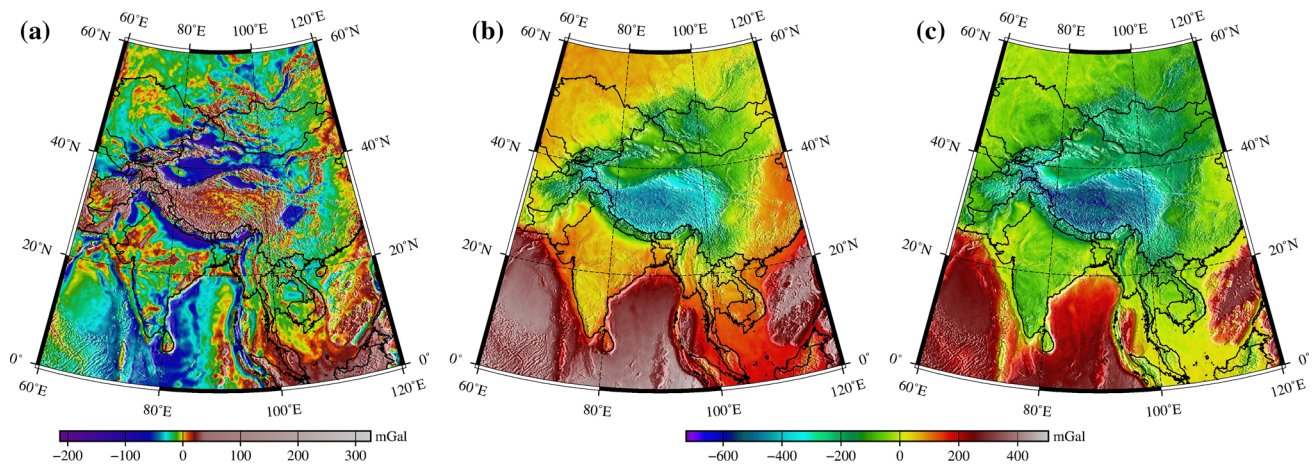


Fig. 13 **a** Gravity disturbance computed with the EIGEN-6C4 global gravity model. **b** Bouguer gravity anomaly computed with the gravity disturbance in (a) and the topographic gravity effect in Fig. 12a. **c** Bouguer gravity anomaly computed with the EIGEN-6C4 model using the ICGEM's service

an arbitrary shaped approximate isometric body and forward modeling its outside gravitational potential, gravity and its gradient tensors. The method could also be easily applied to other applications such as the isostatic corrections.

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A Appendix

In order to map the partition on the icosahedral facets outward to a certain surface, geocentric radiuses of the surface under an equal-angle grid are firstly determined (Fig. 14). As for the terrain surface of the Earth, discussed by Balmino et al. (2011) and Tenzer et al. (2009), given some systematic approximations adopted, the geocentric radius R could be expressed as $R = r + H + N$ without introducing large error, in which r means the geocentric radius of the Earth's reference sphere or ellipsoid, H and N are the corresponding altitude and geoid undulation. If the radius r is latitude-dependent, the Earth's ellipticity can be easily taken into account by $r = ab\sqrt{a^2\sin^2\theta_c + b^2\cos^2\theta_c}$, where θ_c is the geocentric colatitude, a and b are the semimajor and semiminor axes of the Earth's reference ellipsoid, respectively.

After determining geocentric radiuses under the equal angular grid, radiuses of the PGIM's vertices are interpolated using the spherical bilinear interpolation method (Wang et al. 2006) as following. Geocentric radius r_{ave} of a PGIM's vertex m (ϕ_m, θ_m) could be determined by grid points closed to

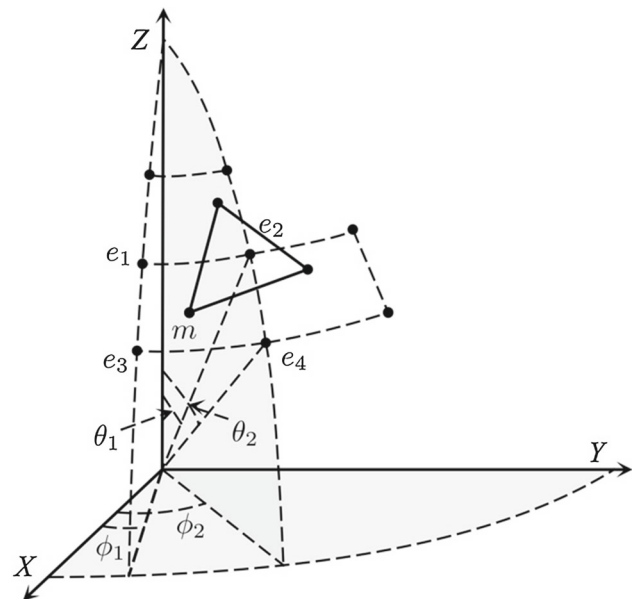


Fig. 14 A diagrammatic sketch of the relationship between the PGIM's vertices and an equal-angle grid. In which, m is a vertex of the PGIM, and e_1 to e_4 are four data points of the equal-angle grid. The geocentric radius of m is interpolated by these four points' radiuses

m . Provided $e_1(\phi_1, \theta_1, r_1)$, $e_2(\phi_2, \theta_2, r_2)$, $e_3(\phi_3, \theta_3, r_3)$ and $e_4(\phi_4, \theta_4, r_4)$ are four data points of the equal angular grid which form the smallest spherical quadrangular region with m in it (Fig. 14). The interpolation can be performed as:

$$r_{ave}(\phi_m, \theta_m) = (A \cos \phi_m + B \theta_m + C \theta_m \cos \phi_m + D) / \Delta \quad (1)$$

$$A = \Delta \theta_1 (r_4 - r_2) + \theta_2 (r_1 - r_3) \quad (2)$$

$$B = (r_1 - r_2) \cos \phi_1 + (r_4 - r_3) \cos \phi_2, \quad (3)$$

$$C = r_3 + r_2 - r_1 - r_4, \quad (4)$$

$$D = \theta_2 \cos \phi_2 r_3 - \theta_2 \cos \phi_1 r_1 + \theta_1 \cos \phi_1 r_2 - \theta_1 \cos \phi_2 r_4, \quad (5)$$

$$\Delta = (\theta_2 - \theta_1) (\cos \phi_2 - \cos \phi_1). \quad (6)$$

Discussed by Du et al. (2012), the interpolation could be taken as an average and error reduction process of the equal angular data set. The error of the interpolation is very low if the resolution of the original grid data is very high. And Tsoulis et al. (2003) pointed out that the resulting surfaces had the geometrical constraint to pass through the exact positions of the original data. Therefore, the error introduced by the interpolation is ignored here.

B Appendix

The exterior gravitational potential, along with its vectors and gradient tensors, of a polyhedron can be forward calculated using the algorithm proposed by Werner and Scheeres (1996). The algorithm is derived in closed form so that there is no error produced during the process of forward calculating theoretically. Assuming the distribution of density inside a polyhedron, expressed as ρ in the following equations, is constant, all facets of the polyhedron are triangles and vertices of a facet are ordered anticlockwise as one observes the facet from outside of the polyhedron. Then the polyhedron's exterior gravitational potential V , gradient ∇V and gradient tensors $\nabla \nabla V$ are given as below, in which G is the gravitational constant. Gradient directions here are x , y and z in the Cartesian coordinate system.

$$V = \frac{1}{2} G \rho \left(\sum_{e=1}^n \vec{r}_e \cdot \mathbf{E}_e \cdot \vec{r}_e \cdot L_e + \sum_{f=1}^m \vec{r}_f \cdot \mathbf{F}_f \cdot \vec{r}_f \cdot \omega_f \right), \quad (7)$$

$$\nabla V = -G \rho \left(\sum_{e=1}^n \mathbf{E}_e \cdot \vec{r}_e \cdot L_e + \sum_{f=1}^m \mathbf{F}_f \cdot \vec{r}_f \cdot \omega_f \right), \quad (8)$$

$$\nabla \nabla V = -G \rho \left(\sum_{e=1}^n \mathbf{E}_e \cdot L_e + \sum_{f=1}^m \mathbf{F}_f \cdot \omega_f \right). \quad (9)$$

in which m and n represent number of facets and edges of the polyhedron, respectively. This feature indicates that the exterior gravitational fields of a polyhedron could be expressed as a sum of effect of all its facets and edges. For instance, as the sketch of a polyhedron shown in Fig. 15, $\vec{r}_f$ and $\vec{r}_e$ stand for two vectors from the observation point P to a triangular facet of the polyhedron, ABC , and one of its edges, CA . Meanwhile, $\mathbf{F}_f$ and $\mathbf{E}_e$ represent tensor products of facet and edge accordingly. ω_f and L_e are two coefficients decided by the relative position between the observing point P and the face and edge. The expressions mentioned above are given by

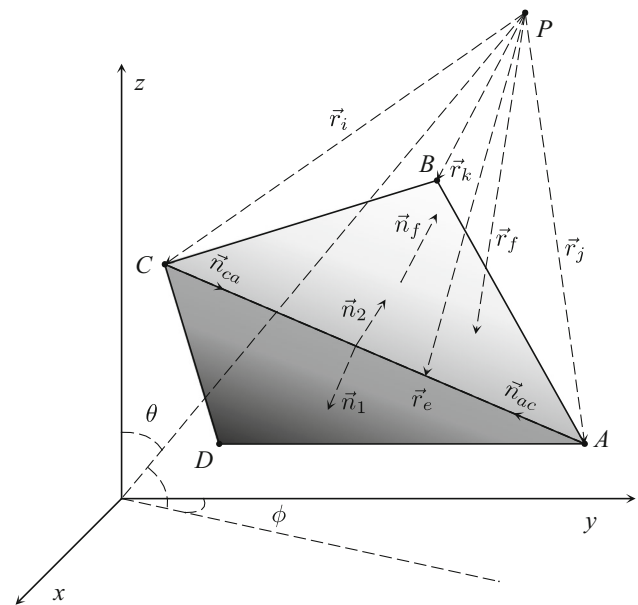


Fig. 15 A diagrammatic sketch of a tetrahedron $ABCD$ placed in the Cartesian coordinate. P is the observation point, $\vec{r}$ represents vector, and $\vec{n}$ is a normal vector

$$\mathbf{E}_e = \hat{n}_1 \hat{n}_{ca} + \hat{n}_2 \hat{n}_{ac}, \quad (10)$$

$$\mathbf{F}_f = \hat{n}_f \hat{n}_f, \quad (11)$$

$$L_e = \frac{\vec{r}_i + \vec{r}_j + d_{ij}}{r_i + r_j - d_{ij}}, \quad (12)$$

$$\omega_f = 2 \tan^{-1} \frac{\vec{r}_i \cdot \vec{r}_j \times \vec{r}_k}{\vec{r}_i \vec{r}_j \vec{r}_k + \vec{r}_i (\vec{r}_j \cdot \vec{r}_k) + \vec{r}_j (\vec{r}_k \cdot \vec{r}_i) + \vec{r}_k (\vec{r}_i \cdot \vec{r}_j)}. \quad (13)$$

In equations above, $\vec{r}_i$, $\vec{r}_j$ and $\vec{r}_k$ represent three vectors from P to the face's vertices C , A and B , with its module $\vec{r}_i$, $\vec{r}_j$ and $\vec{r}_k$, respectively. $\vec{n}_f$ stands for the outside normal vector of the face ABC . It equals the normalized vector of $\vec{AB} \times \vec{BC}$. d_{ij} represents the distance between the ends of vectors $\vec{r}_i$ and $\vec{r}_j$ which equals the length of edge CA here. Moreover, $\vec{n}_1$ and $\vec{n}_2$ are two outside normal vectors of edge CA that lie on the face ABC and face ACD , respectively. These two vectors could be obtained by the cross-product between a face's outside normal vector and one of its edge's normal vector accordingly. For instance, $\vec{n}_1$ equals $\vec{n}_f \times \vec{n}_{ca}$. Expressions such as $\mathbf{F}_f = \hat{n}_f \hat{n}_f$ means the tensor product of two normal vectors which is a 3×3 matrix. For each facet of the polyhedron, it has a $\mathbf{F}_f$ and ω_f associated with. And each edge has an individual $\mathbf{E}_e$ and L_e . Subsequently, the gravitational field of the polyhedron could be obtained by calculating the sum of effects of all its facets and edges.

As all computations are conducted in the Cartesian coordinate, the outcome gravitational fields are with respect to

the Cartesian coordinate as well. A coordinate transformation matrix $\mathbf{R}$ is added into Eqs. (8) and (9), as we need to forward calculate the gravitational fields of a polyhedron in directions reference to the spherical coordinate. Given the position of the observation point P in spherical coordinates $(\phi_P, \theta_P, \bar{r}_P)$, then $\mathbf{R}$ is given by

$$\mathbf{R} = \begin{vmatrix} \sin \theta_P \cos \phi_P & \sin \theta_P \sin \phi_P & \cos \theta_P \\ \cos \theta_P \cos \phi_P & \cos \theta_P \sin \phi_P & -\sin \theta_P \\ -\sin \phi_P & \cos \phi_P & 0 \end{vmatrix}. \quad (14)$$

Subsequently, gradient and gradient tensors of a polyhedron's gravitational potential oriented in the directions ϕ , θ and r are expressed as below.

$$\nabla V = -G\rho \left(\sum_{e=1}^n \mathbf{E}_e \cdot \bar{\mathbf{r}}_e \cdot \mathbf{R}^T \cdot L_e + \sum_{f=1}^m \mathbf{F}_f \cdot \bar{\mathbf{r}}_f \cdot \mathbf{R}^T \cdot \omega_f \right), \quad (15)$$

$$\nabla \nabla V = -G\rho \left(\sum_{e=1}^n \mathbf{R} \cdot \mathbf{E}_e \cdot \mathbf{R}^T \cdot L_e + \sum_{f=1}^m \mathbf{R} \cdot \mathbf{F}_f \cdot \mathbf{R}^T \cdot \omega_f \right). \quad (16)$$

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